

Room 10 (P4 D-algebraicity): moderator summary

2026-09-26

One seat (Hilbert) attempted to construct the missing disc-analog Nevanlinna growth theorem (ingredient (b) of `P4_note.tex`) from first principles, since neither of the two prior rooms on P4 (Room 4, and Track A4 before it) had attempted building this machinery directly.

Hilbert's derivation. Re-examining the classical proof of Nevanlinna's logarithmic-derivative lemma (Poisson–Jensen, two radii $r < R$), the proof itself is r/R -local and never actually uses $R \rightarrow \infty$ – the plane-vs-disc distinction only enters afterward, in how $R(r)$ is chosen and the growth scale used. Transplanting mechanically ($r \rightarrow \infty \leftrightarrow r \rightarrow 1^-$) gives a candidate disc log-derivative lemma and, combined with the fact that polynomial ODE coefficients have $T(r, \text{coeff}) = O(1)$ on a compact disc (rather than growing like $\log r$ on the plane), yields a candidate bound $T(r, y) = O(\log(1/(1-r)))$ for any y meromorphic on the disc solving a fixed algebraic ODE of order k , degree $d \geq 2$. This is a real, if unreviewed, piece of original derivation.

The bookkeeping question, resolved. The comparison against U 's already-established $n(r, U) = \Theta(1/(1-r))$ pole count (Room 4) hinges entirely on which counting-function convention is standard in disc Nevanlinna theory: $N(r, f) = \int_0^r n(t) dt/t$ (the plane integral, restricted to $r < 1$) or $N(r, f) = \int_0^r n(t) dt/(1-t)$ (a disc-native weighting that would make the boundary play the role ∞ plays in the plane theory). Hilbert flagged this as unresolved and UNVERIFIED-CITATION, without library access. **Checked here against primary sources (web search, cross-referencing the standard definition and Tsuji's disc extension): the standard convention is $N(r, a) = \int_0^r [n(t, a) - n(0, a)]/t dt + n(0, a) \log r$ – the SAME dt/t weighting as the plane theory, not a $dt/(1-t)$ disc-native one.** This is Hilbert's “alternative convention,” which the seat's own analysis already correctly identified as giving an **inconclusive** comparison (same order, no violation) rather than the “disc-native” convention that would have closed P4.

Room verdict: NO PROGRESS, but a real sub-question genuinely resolved. The candidate disc growth theorem, even if it survives full scrutiny, does *not* close P4 under the standard convention – the comparison is inconclusive, exactly as one of Hilbert’s two branches anticipated. This settles a real ambiguity (rather than leaving it open) and rules out a specific hoped-for path to closure. P4 stands exactly where `P4_note.tex`/Room 4 left it: OPEN, with the correct theorem family named, ingredient (a) now precisely known ($O(1/(1 - r))$, not superpolynomial), and ingredient (b)’s standard-convention version now known not to be sufficient even if constructed. The candidate log-derivative-lemma transplant itself (genuinely new, unreviewed) is left in `room10_seat_hilbert.tex` for anyone who wants to pursue a non-standard/alternative growth measure in the future, though that would need independent justification for why it’s the right notion of “growth” to use.